MSE 2 – CASE STUDY REPORT

E-Waste Management, Recycling and Circular Economy

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Case Study 1: Why Japan Recycles More Than Most Countries: Lessons for India

Executive Summary

Japan's recycling approach is built around separating materials at the point of disposal, municipal collection rules, defined roles for local authorities and recyclers, and producer-side obligations for selected product streams. For packaging, the national system connects households, municipalities, a coordinating association and contracted recyclers. The case lesson for India is that rules work best when citizens know how to sort, collection schedules are dependable, and downstream recycling markets and accountability are established. This case concerns packaging and broader resource-recycling arrangements; it should not be read as proof that every Japanese waste stream has a uniformly high recycling rate.

1. Background

Japan faced pressure from high volumes of household packaging waste and limited landfill capacity. The Japan Containers and Packaging Recycling Act was enacted in 1995; packaging represented about 60% of household waste by volume in the cited historical context. The system assigns sorting responsibilities to consumers and collection responsibilities to municipalities, while recycling businesses process eligible sorted materials [R1, R2].

2. Strategy / Initiative

- Households separate eligible containers and packaging according to local municipal instructions; rinsing and sorting reduce contamination.
- Municipalities collect material that meets sorting standards and deliver it through the statutory system.
- A coordinating association arranges transport/recycling contracts and recycler selection; recyclers recover materials and sell recycled outputs.
- Producer responsibility and material-specific laws create a shared chain rather than relying only on disposal-stage action.

3. Impact & Results with Data Table

The cited association explains the institutional flow and historical policy rationale, but the consulted pages do not provide a single current national recycling percentage for all waste streams. Avoid comparing a packaging-specific measure with total municipal waste without checking definitions.

Indicator	Value / finding	Year / scope	Source / qualification
Packaging share in household waste (historical context)	About 60% by volume	1995 context	Japan Containers and Packaging Recycling Association [R1]
Containers and Packaging Recycling Act	Enacted	1995	JCPRA [R1]
Operational chain	Consumers → municipalities → association/recyclers	Current process description	JCPRA [R2]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Clear sorting roles; established municipal collection; dedicated recycling chain.
Weaknesses	Sorting complexity varies by municipality; contamination and participation require ongoing effort.
Opportunities	Improve reuse/refill, design for recyclability, and transparent material-specific reporting.
Threats	Changing packaging materials, low-value mixed plastics, and consumer fatigue can undermine recovery.

5. Comparison with India

India's E-Waste (Management) Rules, 2022 create a national EPR and registration framework, while urban collection is shaped by state/local capacity and a large informal recovery sector [R3, R4]. Japan's packaging model highlights disciplined source separation and dependable downstream hand-off. Adaptation in India would require locally understandable sorting instructions, reliable collection, formal links to authorized processors, and inclusion of existing waste workers—not a simple copy of Japan's municipal system.

6. Recommendations

- Standardize simple, multilingual sorting instructions for priority waste streams and communicate them through housing societies, schools and retailers.
- Publish city-level collection schedules and missed-pickup reporting.
- Create traceable handover records from local collection points to authorized processors.
- Pair producer financing with independent audits of actual recovered material and contamination rates.

7. Conclusion

Japan's transferable lesson is system coordination: household sorting, municipal logistics, producer obligations and qualified recyclers must operate as one chain. India can adapt the principles while designing for local infrastructure and informal-sector realities.

8. References

- [R1] Japan Containers and Packaging Recycling Association, "Point of Act," https://www.jcpra.or.jp/english/act.html
- [R2] Japan Containers and Packaging Recycling Association, "Recycling Process," https://www.jcpra.or.jp/english/process.html
- [R3] CPCB, E-Waste (Management) Rules, 2022 FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf

Case Study 2: Switzerland's E-Waste Success Story

Executive Summary

Switzerland demonstrates a long-running, industry-supported take-back system for electrical and electronic equipment. Advance recycling contributions (ARCs) are included in product prices and finance collection, logistics, recycling and quality assurance. Swico and SENS operate complementary systems with collection points and approved recycling partners. The system's key design features are convenient returns, dedicated financing, broad participation by manufacturers/importers and controlled downstream processing.

1. Background

E-waste requires specialized handling and contains both recoverable materials and potentially hazardous components. Switzerland's systems developed over decades: Swico reports operating since 1994, and the SENS/Swico systems collectively process substantial annual volumes. The Federal Office for the Environment reports that more than half of municipal solid waste is separately collected for recycling, although this municipal-waste indicator is distinct from e-waste performance [R4, R5].

2. Strategy / Initiative

- Consumers pay an advance recycling contribution when purchasing covered products; disposal later does not require a separate fee under the system.
- Manufacturers/importers, retailers, collection points, municipalities and recycling partners participate in take-back.
- Swico and SENS use collection networks and professional recycling partners, with quality controls and dedicated funds.
- The systems publish operational data and adjust fees as costs change.

3. Impact & Results with Data Table

In 2024, Swico and SENS together processed about 135,200 tonnes of WEEE, 2% more than 2023. SENS separately reports 100,762 tonnes processed in 2024, equal to 11.14 kg per resident; it had 1,469 manufacturers/importers subject to ARFs and 442 official collection points in 2024 [R5, R6].

Indicator	Value / finding	Year / scope	Source / qualification
Swico + SENS WEEE processed	~135,200 tonnes	2024	SENS technical report [R5]
SENS system processed	100,762 tonnes	2024	SENS annual figures [R6]
SENS processed per resident	11.14 kg	2024	SENS [R6]
SENS manufacturers/importers subject to ARF	1,469	2024	SENS [R6]
SENS official collection points	442	2024	SENS [R6]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Convenient take-back; earmarked financing; industry participation; specialized processing and reporting.
Weaknesses	System depends on sustained governance and fee adequacy; multiple schemes require coordination.
Opportunities	Expand repair, reuse, refurbishment and recovery of critical materials.
Threats	Rising processing costs, product complexity and exports/flows outside the controlled system.

5. Comparison with India

India's rules also assign EPR obligations to producers and require registration and use of registered recyclers, but the Indian system must operate across a much larger and more diverse geography, with uneven collection access and informal collection channels [R3]. Switzerland's pre-paid contribution and dense take-back network provide a financing/access model to consider, while India needs safeguards to ensure producer payments translate into verified recycling.

6. Recommendations

- Pilot visible, no-hassle e-waste return points at electronics retailers, campuses and municipal facilities.
- Explore transparent product-level or pooled producer financing that covers collection and safe processing.
- Publish annual collection, recycling, reuse and residue data with consistent definitions.
- Build authorized recycler capacity and connect informal collectors through safe, paid aggregation arrangements.

7. Conclusion

Switzerland's experience shows that financing, convenient collection and verified processing reinforce one another. India can adapt these elements through phased pilots and strong monitoring.

8. References

- [R4] Swiss Federal Office for the Environment indicator, "Recycling rate," https://www.indikatoren.admin.ch/public/v2/detail?ind=AB031&lng=en
- [R5] SENS eRecycling, "Technical Report 2025," https://www.erecycling.ch/en/privatpersonen/blog/fachbericht-2025.html
- [R6] SENS eRecycling, 2024 annual report facts and figures, https://www.erecycling.ch/en/ueber-sens/medien-und-publikationen/geschaeftsbericht/overview-2024/facts-figures.html
- [R7] Swico Recycling, "Facts and figures," https://www.swico.ch/en/recycling/basics/facts-and-figures/

Case Study 3: India's E-Waste Management Rules 2022

Executive Summary

The E-Waste (Management) Rules, 2022 shifted India's regulatory approach toward a centralized online EPR system. Producers must meet recycling obligations through registered recyclers, with certificates generated on the CPCB portal based on recycling outputs. The rules also cover registration of manufacturers, producers, refurbishers and recyclers, reporting, audits and environmental compensation. They came into force on 1 April 2023, according to CPCB's FAQ [R3].

1. Background

Electronic products are discarded as they are replaced, repaired or become obsolete. Safe management matters because e-waste contains valuable materials and components requiring controlled handling. India's rules operate under the Environment (Protection) Act framework and apply to listed electrical/electronic equipment and relevant stakeholders [R3].

2. Strategy / Initiative

- Registration: manufacturers, producers, refurbishers and recyclers register through the CPCB portal.
- Producer EPR: producers meet recycling targets through certificates issued for recycling by registered recyclers.
- Material accounting: certificates are linked to recycled end products such as gold, copper, aluminium and iron, as described in the CPCB FAQ.
- Refurbishment: the rules recognize refurbishment certificates and account for extended product life.
- Compliance: quarterly/annual returns, stakeholder audits and environmental compensation support monitoring and enforcement.

3. Impact & Results with Data Table

The framework creates a formal mechanism for tracking producer obligations and recycler outputs. Actual environmental performance depends on collection, accurate reporting, verified processing, capacity and enforcement. The CPCB portal provides registration, target, returns and compliance functions; policy design should be distinguished from measured outcomes [R3, R8].

Indicator	Value / finding	Year / scope	Source / qualification
Rules notified	2 November 2022	2022	MoEFCC listing [R8]
Rules effective	1 April 2023	2023	CPCB FAQ [R3]
Equipment coverage	106 EEE items across 7 categories	As stated in CPCB FAQ	CPCB [R3]
EPR fulfillment route	Registered recyclers and portal-based certificates	Rules framework	CPCB [R3]
Reporting	Quarterly and annual returns; audits	Rules framework	CPCB [R3]

4. SWOT Analysis

SWOT element	Analysis
Strengths	National framework; digital registration/certificate system; targets; audits and environmental compensation.
Weaknesses	Formal portal records alone do not guarantee complete collection or safe handling of all generated e-waste.
Opportunities	Improve traceability, consumer returns, repair/refurbishment and integration of informal collectors.
Threats	False documentation, weak enforcement, unregistered processing and data gaps can reduce trust and outcomes.

5. Comparison with India

Compared with decentralized city programs, the 2022 Rules establish national responsibilities and digital reporting. Implementation remains dependent on state pollution control boards, local collection infrastructure, producer participation and authorized recyclers. International models such as Switzerland illustrate how take-back convenience and dedicated funding can complement India's regulatory EPR architecture [R5, R6].

6. Recommendations

- Strengthen independent verification linking EPR certificates to physical input, process outputs and residue management.
- Expand accessible collection and take-back points, especially outside major metros.
- Make public dashboards show targets, certificates, actual recycled quantities and enforcement outcomes in comparable units.
- Create safe formal partnerships with waste pickers and aggregators, including training and protective equipment.

7. Conclusion

The 2022 Rules provide an important governance structure. Their success should be judged not only by registrations or certificates but by verified collection, safe treatment, material recovery and worker protection.

8. References

- [R3] CPCB, FAQ: E-Waste (Management) Rules, 2022, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- [R8] Ministry of Environment, Forest and Climate Change, Rules and Regulations listing, https://moef.gov.in/rules-regulations-3
- [R9] CPCB E-Waste Management System portal, https://eprewaste.cpcb.gov.in/

Case Study 4: India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

Executive Summary

EPR makes producers responsible for managing products after use. India's e-waste regime uses registration, recycling targets and portal-issued certificates. The central evaluation question is whether documented compliance corresponds to actual collection and environmentally sound recycling. A recent secondary analysis reports a sharp variation in target achievement between FY 2023–24 and FY 2024–25; this is an author-reported assessment, not a substitute for the underlying CPCB dataset. It highlights why performance claims should be checked against definitions, years and verified physical flows [R10].

1. Background

EPR is intended to shift some end-of-life responsibility from municipalities and the public to producers. In e-waste, a certificate-based system can provide accountability, but it may underperform if products are not collected, informal processing remains untracked, records are inaccurate, or recycling capacity is inadequate. CPCB's portal requires registration, targets and returns and warns that non-compliance may attract action [R9].

2. Strategy / Initiative

- Use producer obligations and recycling certificates to finance and document recycling.
- Register producers, recyclers, refurbishers and manufacturers and file periodic returns.
- Assess EPR through an outcome chain: products placed on market → waste collected → verified processing → recovered materials/residues → certificates retired.
- Audit physical material flows and reconcile producer claims with recycler records and downstream sales.

3. Impact & Results with Data Table

A 2026 paper reports EPR target achievement of 92.15% for FY 2023–24 and 48.32% for FY 2024–25, and discusses low recovery for some precious metals. Because these figures are from a secondary research paper, they should be presented as that paper's analysis and cross-checked against official data before being treated as definitive [R10].

Indicator	Value / finding	Year / scope	Source / qualification
Reported EPR achievement	92.15%	FY 2023–24	Secondary analysis [R10]; verify against CPCB source data
Reported EPR achievement	48.32%	FY 2024–25	Secondary analysis [R10]; verify against CPCB source data
System requirement	Registered recycler-based EPR certificates	Rules framework	CPCB FAQ [R3]
Compliance information	Quarterly/annual returns and portal notices	Current portal framework	CPCB portal [R9]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Producer responsibility; digital records; formal recycler channel; enforcement provisions.
Weaknesses	Risk of equating certificate transactions with complete, verified collection and environmental outcomes.
Opportunities	Material-level audits, public reporting, collection partnerships and better incentives for difficult-to-recover materials.
Threats	Fraudulent claims, market concentration, weak traceability and low-value waste streams being neglected.

5. Comparison with India

Switzerland's ARF-funded take-back model makes financing visible and supports collection logistics, while India emphasizes target and certificate compliance through a central portal [R5, R7]. The comparison suggests that compliance architecture needs operational collection systems and transparent performance data. India's scale and informal sector require locally adapted integration rather than a direct institutional transplant.

6. Recommendations

- Publish independently audited mass-balance reports: collected input, recycled output, recovered materials, rejects and final destinations.
- Disclose target achievement alongside collection coverage and verified physical quantities.
- Conduct risk-based audits of certificates, recycler capacity, invoices and downstream material sales.
- Reward producer programs that improve accessible collection, repair/reuse and safe inclusion of informal workers.

7. Conclusion

EPR should be evaluated as an environmental outcome system, not merely a paperwork process. Digital certificates are useful only when supported by traceable material flows and credible verification.

8. References

- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- [R9] CPCB E-Waste Management System, https://eprewaste.cpcb.gov.in/
- [R10] Sharma, A.K. & Pandey, R. (2026), "From Targets to Outcome: An Assessment of EPR Implementation in India's E-Waste Sector," IJFMR, https://www.ijfmr.com/research-paper.php?id=72042

Case Study 5: Bengaluru—India's E-Waste Capital?

Executive Summary

Bengaluru is a major technology and electronics-consumption centre, making it an important setting for studying e-waste generation, collection and recycling. The case title poses a question rather than establishing a verified national ranking. The report therefore treats Bengaluru as a city-level case of e-waste governance and avoids claiming it is definitively India's largest generator without a comparable official dataset. Key issues to examine are institutional and household disposal, bulk consumers, collection access, authorized processing and informal recovery.

1. Background

Rapid turnover of computers, phones, peripherals and household appliances creates end-of-life electronics. Bengaluru's technology economy and urban population make collection systems relevant for households, offices, institutions and IT companies. City-level generation estimates may differ according to whether they count waste generated, collected, formally recycled or handled by informal operators; comparisons require consistent definitions.

2. Strategy / Initiative

- Map e-waste sources: households, IT offices, institutions, retailers and repair markets.
- Use producer take-back, authorized collection points and registered recyclers under the national rules.
- Provide secure data-wiping or destruction documentation for computers and storage devices, which can encourage responsible return.
- Coordinate municipal authorities, Karnataka pollution-control authorities, producers, recyclers, resident groups and informal collectors.

3. Impact & Results with Data Table

The sources consulted for this report did not provide a current, comparable official Bengaluru-only e-waste generation and recycling series. This is a data gap, not evidence that the city has no activity. A defensible city case should add the latest municipal/state inventory and distinguish estimated generation from verified formal processing.

Indicator	Value / finding	Year / scope	Source / qualification
City-specific annual e-waste generated	Not established in consulted sources	Use latest official inventory	Do not substitute national estimates
Formal recycling quantity	Not established in consulted sources	Request state/authorized recycler data	Define reporting boundary
Relevant national rules	E-Waste (Management) Rules, 2022	Effective 1 April 2023	CPCB FAQ [R3]
Compliance platform	CPCB EPR portal	Ongoing	CPCB [R9]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Concentrated institutional and technology-sector users can support organized collection and bulk-return programs.
Weaknesses	City-level data may be fragmented across producers, local bodies, repairers and recyclers.
Opportunities	Corporate take-back drives, secure IT asset disposition, campus collection and digital tracking.
Threats	Informal dismantling without safeguards, data-security concerns and leakage to unregistered channels.

5. Comparison with India

Compared with Kerala's state-supported coordination or Pune's worker-integrated municipal service model, Bengaluru's case can focus on bulk institutional generators and technology-sector procurement/retirement systems. India's national EPR rules

provide the compliance framework, but city implementation needs collection access, local coordination and verified records [R3, R11].

6. Recommendations

- Prepare a city e-waste baseline with separate categories for generation, collection, reuse, formal recycling and residual disposal.
- Require public institutions and large IT users to use documented authorized take-back and data sanitization.
- Set up convenient drop-off/retailer return points and publish locations and accepted items.
- Create partnerships that bring informal collectors into safe aggregation and referral pathways.

7. Conclusion

Bengaluru is a useful case for studying urban e-waste governance, but the “capital” label requires a comparable, current dataset. Reliable measurement should precede rankings and policy claims.

8. References

- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- [R9] CPCB E-Waste Management System, https://eprewaste.cpcb.gov.in/
- [R11] Global E-waste Monitor 2024, ITU/UNITAR, https://www.itu.int/en/ITU-D/Environment/Pages/Publications/The-Global-E-waste-Monitor-2024.aspx

Case Study 6: Mumbai's E-Waste Ecosystem

Executive Summary

Mumbai's e-waste ecosystem includes consumers, offices, repair and resale markets, collectors, dismantlers, recyclers and regulatory authorities. Research on Maharashtra's authorized e-waste sector identifies awareness of environmental impacts as a major collection barrier and data security as an important driver of consumer trust. This suggests that collection programs need both public awareness and credible data-handling assurances. Mumbai's broader informal recovery economy is economically significant but needs safe, traceable links to formal processing [R12].

1. Background

Mumbai is a dense metropolitan market with households, commercial offices and electronic equipment users. E-waste collection is influenced by the value of recoverable materials, convenience, resale/repair opportunities and confidence that personal data will be protected. A 2020 study based on a Mumbai recycling-company case and questionnaire-based investigation of processing units examined collection drivers, barriers and handling practices [R12].

2. Strategy / Initiative

- Combine producer take-back and authorized recycling with accessible collection points.
- Offer secure data destruction or certified wiping for computers, phones and storage devices.
- Provide awareness about environmental and health risks and where to return devices.
- Track material through collectors, aggregators, dismantlers and registered recyclers; train workers in safe handling.

3. Impact & Results with Data Table

The Mumbai-focused study found that lack of awareness of environmental impacts was the greatest collection constraint in its qualitative analysis, while data security was the most crucial driver for increasing collection. It also reported that quantity/type of e-waste mattered more than distance to the processing unit in the studied context [R12]. These are study findings, not universal city-wide percentages.

Indicator	Value / finding	Year / scope	Source / qualification
Main collection barrier identified	Low awareness of environmental impacts	Study published 2020	Maharashtra/Mumbai case study [R12]
Key collection driver	Data security / trust	Study published 2020	[R12]
Factors shaping collection	Quantity and type more important than distance in study	Study context	[R12]
National regulatory route	Registered recycler-based EPR	Rules effective 2023	CPCB [R3]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Dense consumer/commercial base; existing repair, resale and collection networks; authorized facilities in Maharashtra.
Weaknesses	Fragmented collection; trust and awareness gaps; uneven traceability across channels.
Opportunities	Secure corporate IT asset return, retailer take-back, formal-informal partnerships and data dashboards.
Threats	Unsafe dismantling, hazardous exposure, data misuse and diversion from authorized processing.

5. Comparison with India

Mumbai illustrates a metropolitan ecosystem where private collection and informal recovery interact with formal regulation. Kerala's CKCL model provides an example of state/local-government coordination, while Pune's SWaCH demonstrates how waste-worker organizations can deliver front-end services [R13, R14]. Mumbai can combine these approaches with secure data services for institutional e-waste.

6. Recommendations

- Make data wiping/destruction certificates standard for institutional collection campaigns.
- Run public awareness campaigns explaining hazards, reuse options and authorized return channels.
- Register and train aggregators; provide PPE and safe storage guidance with referral links to authorized recyclers.
- Publish ward-level collection access and verified processing data.

7. Conclusion

Mumbai's case shows that responsible e-waste collection depends on trust as much as infrastructure. Data security, awareness and safe worker participation should be treated as core system features.

8. References

- [R12] Singh, A., Panchal, R. & Naik, M. (2020), "Circular economy potential of e-waste collectors, dismantlers, and recyclers of Maharashtra: a case study," Environmental Science and Pollution Research; abstract at https://pubmed.ncbi.nlm.nih.gov/32291638/
- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- [R13] Clean Kerala Company, https://cleankeralacompany.com/
- [R14] SWaCH Cooperative, https://swachcoop.com/

Case Study 7: Kerala's Clean Kerala Company

Executive Summary

Clean Kerala Company Limited (CKCL), a Government of Kerala undertaking associated with the Local Self Government Department, provides forward linkages for local governments to process and dispose of solid waste. It works with local governments, Haritha Karma Sena, Suchitwa Mission, the Kerala Solid Waste Management Project, the public and private partners. Its work includes segregated plastic, rejects, e-waste, hazardous e-waste and glass [R13, R15].

1. Background

Local governments often need reliable downstream processing and disposal partners after household-level collection and segregation. CKCL acts as a state-level coordinating and service entity, connecting local collection systems with processors and end users. The Kerala LSGD page describes coverage and material flows, including e-waste being transferred to recycling companies approved by pollution-control authorities [R15].

2. Strategy / Initiative

- Coordinate local governments and Haritha Karma Sena for collection of segregated non-biodegradable waste.
- Provide forward linkages for processing/disposal, including sending e-waste to approved recyclers.
- Channel shredded plastic toward polymerized road construction where suitable and send refuse-derived fuel/rejects to cement factories through tie-ups.
- Develop agreements for hazardous e-waste disposal and report collected quantities.

3. Impact & Results with Data Table

CKCL's website dashboard reports 2,678 MT of e-waste collected and 205 MT of hazardous e-waste collected (dashboard values shown on the site at time of access). Kerala LSGD separately reports 2,265 MT e-waste and 161 MT hazardous e-waste collected on its institutional page; these are differing published snapshots and should not be added together. The LSGD page also reports CKCL association with 667 rural and 47 urban local self-government bodies for segregated plastic collection [R13, R15].

Indicator	Value / finding	Year / scope	Source / qualification
E-waste collected	2,678 MT	Website dashboard snapshot	CKCL [R13]
Hazardous e-waste collected	205 MT	Website dashboard snapshot	CKCL [R13]
E-waste collected	2,265 MT	Institutional page snapshot	Kerala LSGD [R15]
Local governments associated for segregated plastic collection	667 rural + 47 urban	LSGD page snapshot	Kerala LSGD [R15]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Government-backed coordination; local-government linkages; partnerships and reported material flows.
Weaknesses	Performance depends on local segregation, logistics, processing capacity and timely public data updates.
Opportunities	Expand collection coverage, traceability, public dashboards and material-specific recovery.
Threats	Market fluctuations, transport costs, contamination and limited downstream outlets for some materials.

5. Comparison with India

CKCL is a state-level coordination example. India's EPR rules place obligations on producers, while CKCL demonstrates the importance of local collection and downstream hand-off. Pune's SWaCH emphasizes a worker cooperative at the front end; Kerala combines local-government systems with a state company providing forward linkages [R3, R14, R15].

6. Recommendations

- Use one consistent, dated dashboard with definitions for collected, transferred, recycled and disposed quantities.
- Expand public e-waste collection drives through local bodies, schools, offices and retailers.
- Publish recycler authorization and final destination records for e-waste consignments.
- Provide regular training and protective equipment for collection workers.

7. Conclusion

CKCL demonstrates how a state-level institution can connect decentralized local collection with processing and disposal partners. Reliable reporting and safe downstream verification are essential to sustain public confidence.

8. References

- [R13] Clean Kerala Company Limited, official website/dashboard, https://cleankeralacompany.com/
- [R15] Kerala Local Self Government Department, CKCL profile, https://lsgd.kerala.gov.in/en/waste-management/supporting-institutions/ckcl/
- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf

Case Study 8: Pune's SWaCH Model

Executive Summary

SWaCH is a worker-owned cooperative of self-employed waste collectors and other urban poor that provides front-end waste services in Pune. It links household collection and recycling with livelihoods and civic service delivery. Its model recognizes waste pickers as service providers rather than treating them only as an informal problem. SWaCH's published 2025 impact figures report coverage of one million households daily, more than 4,000 waste pickers integrated and 82,891 tonnes recycled annually; these are organization-reported figures [R14].

1. Background

Urban waste systems depend on door-to-door collection, segregation and recovery of recyclable materials. Waste pickers already perform valuable collection and sorting, but often face insecure livelihoods and limited recognition. SWaCH provides an organized cooperative structure and works with Pune's municipal system to deliver services and promote recycling [R14].

2. Strategy / Initiative

- Organize self-employed waste collectors through a cooperative owned by workers.
- Provide front-end household collection and recovery of recyclable materials, supporting source segregation.
- Work with municipal authorities and residents; offer additional services such as V-Collect for reusable/electronic items and composting support.
- Use field visits and public education to explain the city's waste chain and safe disposal.

3. Impact & Results with Data Table

SWaCH's 2025 website reports 1,000,000 households covered daily, 4,000+ waste pickers integrated, 82,891 tonnes recycled annually, ₹113 crore saved to Pune Municipal Corporation, and coverage of over 70% of Pune. These are self-reported program indicators; the reporting period and calculation method should be checked when citing them in formal evaluation [R14].

Indicator	Value / finding	Year / scope	Source / qualification
Households covered daily	1,000,000	SWaCH Impact 2025	SWaCH [R14]
Waste pickers integrated	4,000+	SWaCH Impact 2025	SWaCH [R14]
Waste recycled annually	82,891 tonnes	SWaCH Impact 2025	SWaCH [R14]
Reported savings to PMC	₹113 crore	SWaCH Impact 2025	SWaCH [R14]
City coverage	Over 70% of Pune	Website claim	SWaCH [R14]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Worker ownership; established community interface; livelihood integration; measurable service indicators.
Weaknesses	Service quality and income may depend on household participation, recyclable markets and municipal agreements.
Opportunities	Expand e-waste collection, repair/reuse referrals, composting and worker-led awareness.
Threats	Policy or contract changes, volatile recyclable prices, unsafe waste streams and competition from unorganized intermediaries.

5. Comparison with India

SWaCH offers a worker-centred service model, while CKCL in Kerala offers state-level downstream coordination. India's EPR framework can complement SWaCH by directing e-waste collected through trusted worker networks to registered recyclers, provided workers receive fair compensation and safety support [R3, R13, R14].

6. Recommendations

- Formalize service agreements that protect worker voice, fair compensation and continuity of collection.
- Create safe e-waste sorting/aggregation protocols and connect the cooperative with registered recyclers.
- Publish independently reviewed metrics, including worker earnings, safety, material recovery and service coverage.
- Scale public education on segregation and reuse through resident associations and schools.

7. Conclusion

SWaCH demonstrates that integrating waste pickers can improve service delivery while supporting livelihoods. Its principles are relevant to inclusive urban circular-economy programs.

8. References

- [R14] SWaCH Cooperative, “SWaCH Impact 2025,” https://swachcoop.com/
- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf
- [R13] Clean Kerala Company, https://cleankeralacompany.com/

Case Study 9: Dharavi Recycling Ecosystem

Executive Summary

Dharavi's recycling economy is a dense network of waste pickers, neighborhood collectors, transporters, sorters, brokers, dealers and small remanufacturers. It demonstrates how discarded materials can become inputs for livelihoods and production through decentralized recovery. A published chapter estimates that, at its peak, Dharavi's 13 Compounds recycled 80% of Mumbai's hard waste and supported around 250,000 informal waste workers across the wider network; this is a historical/author estimate and should not be treated as a current audited count [R16].

1. Background

Dharavi's recycling clusters grew around accessible discarded material, specialized labor, trading relationships and low barriers to entry. The system recovers value from materials that might otherwise be landfilled, while workers often operate in constrained conditions and may face occupational, environmental and livelihood insecurity [R16].

2. Strategy / Initiative

- Recover discarded material through a chain of collection, sorting, aggregation, trading and remanufacturing.
- Use specialized knowledge and labor to separate material types and supply secondary raw materials to businesses.
- Operate through interconnected small enterprises and informal market relationships.
- Build circular value by returning recovered materials to production, while recognizing the need for safe working conditions and environmental controls.

3. Impact & Results with Data Table

The cited chapter's estimate—80% of Mumbai's hard waste recycled by the 13 Compounds at its peak, with about 250,000 informal workers across the described wider ecosystem—is a contextual estimate, not a current citywide measurement. No current audited tonnage series was identified in the consulted source [R16].

Indicator	Value / finding	Year / scope	Source / qualification
Share of Mumbai hard waste recycled by 13 Compounds	Estimated 80% at peak	Historical estimate	Springer chapter [R16]
Workers supported in wider informal ecosystem	Around 250,000	Author estimate; historical context	Springer chapter [R16]
Current verified annual tonnage	Not provided in consulted source	Data gap	Needs current field/municipal study

4. SWOT Analysis

SWOT element	Analysis
Strengths	High material recovery; specialized skills; flexible small-enterprise network; livelihood creation.
Weaknesses	Informal status can limit access to finance, social protection, safe equipment and formal market contracts.
Opportunities	Worker-led formal partnerships, safer processing, improved material traceability and value-added recycling.
Threats	Pollution and occupational hazards, unstable scrap prices, displacement pressures and loss of livelihoods during redevelopment.

5. Comparison with India

Dharavi's informal circular economy differs from formal producer-led systems such as Switzerland's take-back networks and India's registered-recycler EPR system. It also complements worker-integration approaches such as Pune's SWaCH. Policy

should preserve the recovery skills and livelihoods while improving safety, environmental performance and traceability [R3, R14, R16].

6. Recommendations

- Recognize worker groups and micro-enterprises as partners in collection and sorting, with registration pathways and practical compliance support.
- Provide shared safe workspaces, ventilation, PPE, fire safety and hazardous-material handling training.
- Create contracts linking aggregators to registered recyclers and document quantities and final destinations.
- Undertake participatory research to establish current employment, tonnage, income and environmental baselines before making claims.

7. Conclusion

Dharavi shows the economic value of informal circularity, but high recovery does not automatically mean safe or equitable recycling. Inclusive upgrading should improve health, environmental controls and livelihoods together.

8. References

- [R16] “Compound 13 Lab, Mumbai: Learning Through Waste Ecologies,” Springer Nature chapter, https://link.springer.com/chapter/10.1007/978-3-031-82338-1_3
- [R14] SWaCH Cooperative, https://swachcoop.com/
- [R3] CPCB, E-Waste Rules FAQ, https://eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf

Case Study 10: Dell's Circular Economy Strategy

Executive Summary

Dell describes circularity across product design, repair/refurbishment, material selection, packaging, recovery services and refurbished-product reuse. Its stated principles include durability, easier repair and upgrades, dematerialization, use of recycled/renewable/lower-emissions materials, design for disassembly and lifecycle services. Dell also states a 2030 product take-back ambition of reusing or recycling one metric ton for every metric ton of product customers buy, and a goal for all packaging to use recycled/renewable or reused material by 2030 [R17, R18].

1. Background

Electronics manufacturing uses materials and energy, while short product lifetimes increase waste and demand for virgin resources. Circular design seeks to keep products and materials in use longer through durability, repair, reuse, refurbishment and recycling. Dell frames this as a lifecycle and supply-chain strategy rather than only an end-of-life recycling program [R17].

2. Strategy / Initiative

- Design products for durability, repair, upgrades and easier disassembly.
- Use recycled, renewable and lower-emissions materials and optimize material use and packaging.
- Offer repair, recovery, recycling and refurbished-product options across product life cycles.
- Partner across suppliers and recovery networks; track long-term goals for take-back and packaging circularity.

3. Impact & Results with Data Table

Dell's public pages describe strategy and 2030 ambitions, but the consulted pages do not provide a complete audited result table for progress against every target. The 2030 statements should therefore be labeled goals, not achieved outcomes. Dell also cites the Global E-waste Monitor's 2022 global estimate of 62 billion kg generated and 22.3% formally collected/recycled, illustrating the scale of the challenge [R17, R18, R19].

Indicator	Value / finding	Year / scope	Source / qualification
Product take-back ambition	1 tonne reused/recycled per tonne sold	By 2030	Dell [R18]
Packaging ambition	100% recycled/renewable material or reused packaging	By 2030	Dell [R18]
Global e-waste generated	62 billion kg	2022	ITU/UNITAR [R19]
Global documented formal recycling	22.3% of generated e-waste	2022	ITU/UNITAR [R19]

4. SWOT Analysis

SWOT element	Analysis
Strengths	Lifecycle design approach; repair/refurbishment; recovery services; measurable future targets.
Weaknesses	Targets depend on customer returns, reverse logistics, repair economics and supplier data.
Opportunities	Modular repair, longer support, refurbished markets and recycled-material sourcing.
Threats	Rapid technology change, complex material mixes, low return rates and green-claim scrutiny.

5. Comparison with India

Dell's corporate strategy is producer-led and global, whereas India's EPR rules create statutory obligations across producers and registered recyclers. Dell's design-for-repair and take-back practices can complement Indian compliance, but effective implementation needs accessible collection, affordable repair, authorized recycling and transparent reporting [R3, R17].

6. Recommendations

- Expand repairability, spare-part availability and refurbishment pathways in markets with high device reuse potential.
- Publish annual progress against take-back and packaging goals with boundaries, methodology and independently verifiable data.
- Design India-specific return options through retailers, service centers, campuses and corporate IT asset programs.
- Ensure collected products are securely data-wiped and routed to reuse or authorized recycling according to condition.

7. Conclusion

Dell's approach shows how circularity can be embedded in design and lifecycle services. For India, product durability, repair, take-back and transparent EPR evidence are complementary rather than competing measures.

8. References

- [R17] Dell India, "Accelerating the Circular Economy," https://www.dell.com/en-in/dt/corporate/social-impact/advancing-sustainability/accelerating-the-circular-economy.htm
- [R18] Dell, "Accelerating the Circular Economy" (take-back and packaging goals), https://www.dell.com/en-au/learn/corporate-impact/lp/circular-economy
- [R19] ITU/UNITAR, Global E-waste Monitor 2024, https://www.itu.int/en/ITU-D/Environment/Pages/Publications/The-Global-E-waste-Monitor-2024.aspx
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- [R18] Dell, “Accelerating the Circular Economy” (take-back and packaging goals), https://www.dell.com/en-au/learn/corporate-impact/lp/circular-economy
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Submission check: confirm the instructor’s preferred citation style; verify any figures against the latest official reports before final submission; keep goals, estimates, organization-reported figures and audited outcomes clearly distinguished.